

HDOC Day-15 Activity

Menu-Driven C Program Using Switch-Case

Question

Write a menu-driven C program using switch-case to input a positive integer and perform one of the following operations based on the user's choice:

- Check whether all digits are even.
- Check whether all digits are prime.

Algorithm

1. Start.
2. Input a positive integer num.
3. If num <= 0, display "Please enter a positive integer" and stop.
4. Display the menu: 1. Check whether all digits are even 2. Check whether all digits are prime.
5. Input the user's choice and copy num into temp.
6. Use switch(choice).
7. For Case 1, set flag = 1. Repeat while temp > 0: digit = temp % 10. If digit % 2 != 0, set flag = 0 and break. Otherwise set temp = temp / 10. Display the result according to flag.
8. For Case 2, set flag = 1. Repeat while temp > 0: digit = temp % 10. If digit is not 2, 3, 5, or 7, set flag = 0 and break. Otherwise set temp = temp / 10. Display the result according to flag.
9. For any other choice, display "Invalid choice".
10. Stop.

C Program

```
#include <stdio.h>

int main()
{
    int num, choice, temp, digit, flag;

    printf("Enter a positive integer: ");
    scanf("%d", &num);

    if (num <= 0)
    {
        printf("Please enter a positive integer\n");
        return 0;
    }

    printf("\nMENU\n");
```

```

printf("1. Check whether all digits are even\n");
printf("2. Check whether all digits are prime\n");
printf("Enter your choice: ");
scanf("%d", &choice);

temp = num;

switch (choice)
{
    case 1:
        flag = 1;
        while (temp > 0)
        {
            digit = temp % 10;
            if (digit % 2 != 0)
            {
                flag = 0;
                break;
            }
            temp = temp / 10;
        }
        if (flag == 1)
            printf("All digits are even\n");
        else
            printf("All digits are not even\n");
        break;

    case 2:
        flag = 1;
        while (temp > 0)
        {
            digit = temp % 10;
            if (digit != 2 && digit != 3 && digit != 5 && digit != 7)
            {
                flag = 0;
                break;
            }
            temp = temp / 10;
        }
        if (flag == 1)
            printf("All digits are prime\n");
        else
            printf("All digits are not prime\n");
        break;

    default:
        printf("Invalid choice\n");
}

return 0;
}

```

Test Case Table

Test Case	Number	Choice	Expected Output
1	24680	1	All digits are even
2	12345	1	All digits are not even
3	2357	2	All digits are prime
4	2379	2	All digits are not prime
5	-25	-	Please enter a positive integer
6	2468	3	Invalid choice

Compilation and Execution

```
(kali㉿kali)-[~/Day15]
$ gcc day15.c -o day15
(kali㉿kali)-[~/Day15]
$
```

Execution - Choice 1: Not all digits even

```
(kali㉿kali)-[~/Day15]
$ ./day15
Enter a positive integer: 12345
1. Check whether all digits are even
2. Check whether all digits are prime
Enter your choice: 1
All digits are not even.
(kali㉿kali)-[~/Day15]
$
```

Execution- Choice 1: All digits even

```
(kali㉿kali)-[~/Day15]
$ ./day15
Enter a positive integer: 24680
1. Check whether all digits are even
2. Check whether all digits are prime
Enter your choice: 1
All digits are even.
(kali㉿kali)-[~/Day15]
$
```

Execution- Choice 2: All digits prime

```
(kali㉿kali)-[~/Day15]
$ ./day15
Enter a positive integer: 2357
1. Check whether all digits are even
2. Check whether all digits are prime
Enter your choice: 2
All digits are prime.
(kali㉿kali)-[~/Day15]
$
```

Execution- Choice 2: Not all digits prime

```
(kali㉿kali)-[~/Day15]
$ ./day15
Enter a positive integer: 2379
1. Check whether all digits are even
2. Check whether all digits are prime
Enter your choice: 2
All digits are not prime.
(kali㉿kali)-[~/Day15]
$
```